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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 7)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The coordinates of the end points of a diameter of a circle are $(5, -2)$ and $(5, 2)$. The length of the radius of the circle is: [1]

(A) ± 2 **(B)** ± 4 **(C)** 4 **(D)** 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) 2

Given: Endpoints of diameter $A(5, -2)$ and $B(5, 2)$.

Formula Used: Diameter $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ and $r = \dfrac{AB}{2}$

$$AB = \sqrt{(5-5)^2 + (2-(-2))^2} = \sqrt{0 + 4^2} = 4 \text{ units}$$

Since radius is a non-negative geometric length:

$$r = \frac{4}{2} = \boxed{\mathbf{2} \text{ units}}$$

Q2. The end points of a diameter of circle are $(2, 4)$ and $(-3, -1)$. The length of its radius is: [1]

(A) $\dfrac{5\sqrt{2}}{2}$ units **(B)** $5\sqrt{2}$ units

(C) $3\sqrt{2}$ units **(D)** $\pm\dfrac{5\sqrt{2}}{2}$ units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\dfrac{5\sqrt{2}}{2}$ units

Given: Endpoints of diameter $A(2, 4)$ and $B(-3, -1)$.

Formula Used: Diameter $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, $r = \dfrac{AB}{2}$

$$AB = \sqrt{(-3-2)^2 + (-1-4)^2} = \sqrt{(-5)^2 + (-5)^2} = \sqrt{25 + 25} = \sqrt{50} = 5\sqrt{2} \text{ units}$$

Radius (always positive):

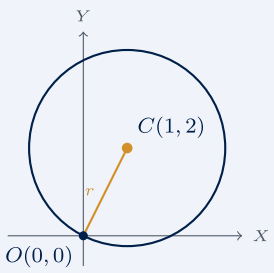
$$r = \frac{5\sqrt{2}}{2} = \boxed{\mathbf{\frac{5\sqrt{2}}{2}} \text{ units}}$$

Q3. In the given figure, a circle is centred at $(1, 2)$. The diameter of the circle is: [1]

(A) 4 **(B)** $2\sqrt{2}$ **(C)** $\sqrt{5}$ **(D)** $2\sqrt{5}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (D) $2\sqrt{5}$



Note: In the standard CBSE Board diagram for this question (omitted in the printed test sheet), the circle passes through the origin $O(0, 0)$.

Given: Centre $C(1, 2)$ and boundary point $O(0, 0)$.

We know the distance and diameter formulas:

Formula Used: $r = OC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, Diameter $= 2r$

$$r = \sqrt{(1-0)^2 + (2-0)^2} = \sqrt{1^2 + 2^2} = \sqrt{1+4} = \sqrt{5} \text{ units}$$

$$\text{Diameter} = 2 \times \sqrt{5} = \boxed{\mathbf{2\sqrt{5}} \text{ units}}$$

Q4. A circle centred at $(-1, 2)$ passes through the point $(0, 3)$. Radius of the circle is: [1]

(A) $2\sqrt{2}$ **(B)** $\sqrt{2}$ **(C)** $\sqrt{26}$ **(D)** 1

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) $\sqrt{2}$

Given: Centre $O(-1, 2)$ and point $P(0, 3)$ on the circle.

Formula Used: $r = OP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$r = \sqrt{(0-(-1))^2 + (3-2)^2} = \sqrt{1^2 + 1^2} = \sqrt{1+1} = \boxed{\mathbf{\sqrt{2}} \text{ units}}$$

Q5. $A(-4, 5)$ and $C(8, 2)$ are the two opposite vertices of a parallelogram $ABCD$. Its diagonals intersect each other at $P(a, b)$. The relation between 'a' and 'b' is: [1]

(A) $b = a - 1.5$ **(B)** $b = a + 1.5$
(C) $b = a - 4.5$ **(D)** $b = a + 4.5$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) $b = a + 1.5$

Given: Opposite vertices $A(-4, 5)$ and $C(8, 2)$. Since diagonals of a parallelogram bisect each other, their intersection $P(a, b)$ is the midpoint of diagonal AC.

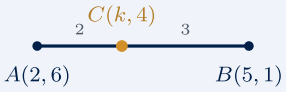
Formula Used: $P(a, b) = \left(\dfrac{x_A + x_C}{2}, \dfrac{y_A + y_C}{2}\right)$

$$a = \frac{-4+8}{2} = \frac{4}{2} = 2, \qquad b = \frac{5+2}{2} = \frac{7}{2} = 3.5$$

Comparing b and a: $b - a = 3.5 - 2 = 1.5 \implies \boxed{\mathbf{b = a + 1.5}}$

Q6. If the point $C(k, 4)$ divides the line segment joining $A(2, 6)$ and $B(5, 1)$ internally in the ratio $2 : 3$, find the value of k. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Section Formula Setup [1 Mark]
Given: Endpoints $A(2, 6)$ and $B(5, 1)$, dividing point $C(k, 4)$, and internal ratio $m_1 : m_2 = 2 : 3$.

We know the internal section formula for the abscissa:

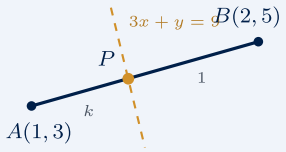
Formula Used: $x = \dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}$

Step 2: Substitution and Simplification [1 Mark]

$$k = \frac{2(5) + 3(2)}{2 + 3} = \frac{10 + 6}{5} = \boxed{\frac{16}{5}}$$

Q7. Determine the ratio in which the line $3x + y - 9 = 0$ divides the line segment joining the points $(1, 3)$ and $(2, 5)$. Find the point of intersection. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Finding the Internal Ratio $k : 1$ [1 Mark]
Given: Endpoints $A(1, 3)$ and $B(2, 5)$, dividing line $3x+y-9 = 0$. Let intersection $P(x, y)$ divide AB internally in ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$

Thus $P = \left(\dfrac{2k + 1}{k + 1}, \dfrac{5k + 3}{k + 1}\right)$. Substituting into $3x + y - 9 = 0$:

$$3\left(\frac{2k + 1}{k + 1}\right) + \left(\frac{5k + 3}{k + 1}\right) - 9 = 0 \implies 3(2k + 1) + (5k + 3) - 9(k + 1) = 0$$

$$6k + 3 + 5k + 3 - 9k - 9 = 0 \implies 2k - 3 = 0 \implies k = \frac{3}{2} \implies \text{Ratio} = \mathbf{3 : 2}$$

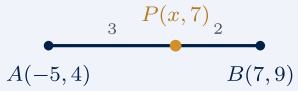
Step 2: Point of Intersection [1 Mark]

$$x = \frac{3(2) + 2(1)}{3 + 2} = \frac{8}{5}, \qquad y = \frac{3(5) + 2(3)}{3 + 2} = \frac{21}{5}$$

$$\boxed{\text{Ratio } = \mathbf{3 : 2} \quad \text{and} \quad P = \left(\frac{\mathbf{8}}{\mathbf{5}}, \frac{\mathbf{21}}{\mathbf{5}}\right)}$$

Q8. A point $P(x, 7)$ divides a line segment joining the points $A(-5, 4)$ and $B(7, 9)$ in a certain ratio. Find the ratio and hence find the value of x. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Finding the Ratio Using y-Coordinate [1 Mark]
Given: Endpoints $A(-5, 4)$ and $B(7, 9)$, dividing point $P(x, 7)$.
Let P divide segment AB in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$

Equating the y-coordinate to 7:

$$\frac{k(9) + 1(4)}{k + 1} = 7 \implies 9k + 4 = 7k + 7 \implies 2k = 3 \implies k = \frac{3}{2} \implies \text{Ratio} = \mathbf{3 : 2}$$

Step 2: Finding Value of x [1 Mark]

$$x = \frac{3(7) + 2(-5)}{3 + 2} = \frac{21 - 10}{5} = \frac{11}{5}$$

$$\boxed{\text{Ratio } = \mathbf{3 : 2} \quad \text{and} \quad \mathbf{x = \frac{11}{5}}}$$

SECTION C: Long Answer & Competency Based Questions (9 Marks)

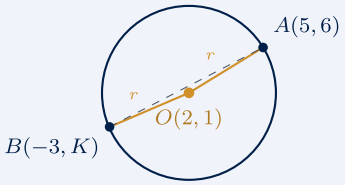
Q9. A circle centered at $(2, 1)$ passes through the points $A(5, 6)$ and $B(-3, K)$. Find the value(s) of K. Hence find length of chord AB. [5]

— OR —

Prove that the point P dividing the line segment joining the points $A(-1, 7)$ and $B(4, -3)$ in the ratio $3 : 2$, lies on the line $x - 3y = -1$. Also find length of PA and PB. [5]

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► SOLUTION (OPTION I): [Total: 5 Marks]



Step 1: Finding Value(s) of K Using Equal Radii $OA = OB$ [2.5 Marks]

Given: Centre $O(2,1)$ passes through $A(5,6)$ and $B(-3,K)$. Since all radii of a circle are equal, $OA = OB \implies OA^2 = OB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$$(5-2)^2 + (6-1)^2 = (-3-2)^2 + (K-1)^2$$

$$3^2 + 5^2 = (-5)^2 + (K-1)^2 \implies 9 + 25 = 25 + (K-1)^2 \implies (K-1)^2 = 9$$

Taking square root on both sides: $K - 1 = \pm 3 \implies K = 1 + 3 = \mathbf{4}$ or $K = 1 - 3 = \mathbf{-2}$.

Step 2: Finding Length of Chord AB for Both Values of K [2.5 Marks]

• **Case 1 ($K = 4$):** $A(5,6)$ and $B(-3,4)$:

$$AB = \sqrt{(-3-5)^2 + (4-6)^2} = \sqrt{(-8)^2 + (-2)^2} = \sqrt{64+4} = \sqrt{68} = \mathbf{2\sqrt{17}} \text{ units}$$

• **Case 2 ($K = -2$):** $A(5,6)$ and $B(-3,-2)$:

$$AB = \sqrt{(-3-5)^2 + (-2-6)^2} = \sqrt{(-8)^2 + (-8)^2} = \sqrt{64+64} = \sqrt{128} = \mathbf{8\sqrt{2}} \text{ units}$$

$$\boxed{K = \mathbf{4} \implies AB = \mathbf{2\sqrt{17}} \text{ units}; \qquad K = \mathbf{-2} \implies AB = \mathbf{8\sqrt{2}} \text{ units}}$$

► SOLUTION (OPTION II): [OR: 5 Marks]



Step 1: Finding P and Checking Line Equation $x - 3y = -1$ [3 Marks]

Given: Endpoints $A(-1,7)$ and $B(4,-3)$, divided internally by P in the ratio $3:2$.

We know the internal section and distance formulas:

Formula Used: $P(x,y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1+m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1+m_2}\right)$

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

$$x = \frac{3(4) + 2(-1)}{3+2} = \frac{12-2}{5} = \frac{10}{5} = 2$$

$$y = \frac{3(-3) + 2(7)}{3+2} = \frac{-9+14}{5} = \frac{5}{5} = 1 \implies P(2,1)$$

Substituting $x = 2, y = 1$ into LHS of $x - 3y = -1$: LHS $= 2 - 3(1) = 2 - 3 = -1 =$ RHS.
Hence, point $P(2,1)$ lies on the line $x - 3y = -1$ **(Proved)**.

Step 2: Calculating Lengths PA and PB [2 Marks]

$$PA = \sqrt{(2-(-1))^2 + (1-7)^2} = \sqrt{3^2 + (-6)^2} = \sqrt{9+36} = \sqrt{45} = \mathbf{3\sqrt{5}} \text{ units}$$

$$PB = \sqrt{(4-2)^2 + (-3-1)^2} = \sqrt{2^2 + (-4)^2} = \sqrt{4+16} = \sqrt{20} = \mathbf{2\sqrt{5}} \text{ units}$$

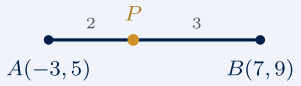
$$\boxed{P(2,1) \text{ lies on } x - 3y = -1; \qquad PA = \mathbf{3\sqrt{5}} \text{ units}, \quad PB = \mathbf{2\sqrt{5}} \text{ units}}$$

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Q10. Competency Case Study: Two cellular transmission towers are situated at points $A(-3,5)$ and $B(7,9)$ on a regional map grid. A booster relay is installed at a point P along the direct cable route AB such that $AP : PB = 2 : 3$. [4]

(i) Find the coordinates of the booster relay P.
(ii) Find the total direct aerial distance between tower A and tower B.
(iii) If a third monitoring station is placed at $C(2,7)$, determine whether station C is equidistant from tower A and tower B.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given: Tower locations $A(-3,5)$ and $B(7,9)$, with booster relay P dividing cable route AB internally in ratio $AP : PB = 2 : 3$. We use the internal section and distance formulas.

We know the internal section and distance formulas:

Formula Used:
$$P(x,y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$$
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

(i) Coordinates of Booster Relay P ($AP : PB = 2 : 3$): [1.5 Marks]

$$x = \frac{2(7) + 3(-3)}{2 + 3} = \frac{14 - 9}{5} = \frac{5}{5} = 1$$
$$y = \frac{2(9) + 3(5)}{2 + 3} = \frac{18 + 15}{5} = \frac{33}{5}$$

Hence, $\mathbf{P} = \left(\mathbf{1}, \frac{\mathbf{33}}{\mathbf{5}}\right)$ (or $(1, 6.6)$).

(ii) Total Direct Aerial Distance Between $A(-3,5)$ and $B(7,9)$: [1 Mark]

$$AB = \sqrt{(7 - (-3))^2 + (9 - 5)^2} = \sqrt{10^2 + 4^2} = \sqrt{100 + 16} = \sqrt{116} = \mathbf{2\sqrt{29}} \text{ units}$$

(iii) Checking Equidistance of $C(2,7)$ from A and B: [1.5 Marks]

$$CA = \sqrt{(-3 - 2)^2 + (5 - 7)^2} = \sqrt{(-5)^2 + (-2)^2} = \sqrt{25 + 4} = \sqrt{\mathbf{29}} \text{ units}$$

$$CB = \sqrt{(7 - 2)^2 + (9 - 7)^2} = \sqrt{5^2 + 2^2} = \sqrt{25 + 4} = \sqrt{\mathbf{29}} \text{ units}$$

Since $CA = CB = \sqrt{29}$ units (indeed $C(2,7)$ is the midpoint of AB), **yes**, station C is equidistant from towers A and B.

$$\boxed{P = \left(\mathbf{1}, \frac{\mathbf{33}}{\mathbf{5}}\right), \quad AB = \mathbf{2\sqrt{29}} \text{ units}, \quad \text{Yes, } CA = CB = \sqrt{29} \text{ units}.}$$

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